

Topology of the real line – 3

Compactness and covering arguments

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Theoretical minimum.

Def. A family of open sets whose union contains K is an *open cover* of K ; the set K is *compact* if every open cover of it has a finite subcover.

Theorem 1 (Heine–Borel and the equivalent forms). For $K \subseteq \mathbb{R}$ the following are equivalent: K is closed and bounded; every open cover of K has a finite subcover; every sequence in K has a subsequence converging in K ; every infinite subset of K has a limit point in K . The nested interval principle is the special case of decreasing closed intervals.

Theorem 2 (Lebesgue number). For every open cover of a compact K there is $\delta > 0$ such that every subset of K of diameter less than δ lies in one member of the cover. Both the statement and the finite subcover property fail on $(0, 1)$.

Fact 1 (Lindelöf and the rising sun). Every cover of a subset of $\mathbb{R}$ by open sets has an at most countable subcover. If f is continuous on $[a, b]$, then $E = \{x : f(y) > f(x) \text{ for some } y > x\}$ is open and on each component (a_k, b_k) of E one has $f(a_k) \leq f(b_k)$; this is F. Riesz's rising sun lemma.

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1. Prove all the equivalences of Theorem 1 and the Lebesgue number lemma, and show by explicit covers that each statement fails for $(0, 1)$ and for $[0, \infty)$.
 2. (Karasev 1.44) Prove that from any cover of an interval by closed segments of non-zero length one can select an at most countable subcover. Show that “non-zero length” cannot be dropped.

3. (Karasev 1.57) Let a smooth $f: \mathbb{R}^n \rightarrow \mathbb{R}$ have zero gradient at every point of a connected compact set X . Prove that f is constant on X . Note that connected and path-connected are different properties, so the naive argument along a path is not available.
4. Prove the rising sun lemma and deduce from it that a continuous function on $[a, b]$ whose set of strict local maxima is dense is constant.
5. (Putnam 1991) Does there exist an infinite sequence of closed discs $D_1, D_2, \dots$ in the plane, with centres $c_1, c_2, \dots$, such that the c_i have no limit point in the finite plane, the sum of the areas of the D_i is finite, and every line in the plane intersects at least one D_i ?